UK CanSat Competition

Team Name : CERC

School : Cambourne Village College

Critical Design Review

Date: 30/01/26

1 Introduction

1.1 Team Organisation and Roles

Team Lead - Organises the members, tasks and building of the Cansat, ensures that deadlines are being met and completes the majority of documentation. Communicates with our mentor on progress and when assistance is needed

Software design lead - Creates and manages software as per required for the CanSat, working closely with the electrical design lead.

Electrical design lead - Designs the necessary circuitry and uses relevant microcontrollers and sensors to create necessary systems for the Cansat.

Mechanical design lead - Uses CAD, sketches and other relevant design tools to engineer the Cansat's structure. Works closely with the electrical design lead.

Landing and ground systems lead - Ensures the reliability of the ground telecommunication systems and designs the landing system.

Testing lead - Designs experiments to test components of and the Cansat as a whole, carries these out and records test results to allow other members to make improvements on the CanSat.

1.2 Mission Overview

1.2.1 Mission Objectives

Our main objectives are :

- To measure air pressure and temperature (primary mission)
- To calculate altitude from air pressure in order to show correlations with other values
- To record the aerodynamic behaviour of our CanSat
- To send measurements to the ground station for later analysis

By measuring values such as temperature and turbulence and comparing them to each other or with altitude, it allows us to scientifically investigate links between these measurements and may reveal correlations between these values

Primary Mission:

Our CanSat's primary mission is:

- To measure air pressure and temperature
- To transmit this data to the ground station every second

We will also calculate altitude directly from our measurement of air pressure.
Air pressure and temperature will be measured with a BMP-280

Data will be transmitted with a Adafruit LoRa RFM9x combined with a Yagi antenna. Transmitted data will be formatted into a CSV file so that data can easily be analysed after the launch. All sensors will be programmed with CircuitPython due to our team's experience in this and its compatibility with many of our chosen sensors

These measurements are scientifically relevant as they allow us to investigate correlation with relevant factors such as temperature and altitude. More detail on this is given in the secondary mission.

More data on choice of sensors and how data is analysed is given in the software and electrical design sections of the CDR.

Secondary Mission:

This part should contain a concise list/description of the secondary mission you are planning to achieve.

Our secondary mission is to measure the turbulence and stability of our CanSat, so that they can be compared with our measurements of temperature, air pressure and altitude.

This is scientifically relevant as it allows us to analyse the effects of temperature and air pressure on our parachute's stability and the turbulence, providing insight into the aerodynamic behaviour of small systems such as a CanSat. Turbulence calculations are crucial in physics and engineering, linking to a wide range of areas such as weather modeling, fluid simulation and aerodynamics. By choosing to measure turbulence inside our CanSat, it opens up the opportunity to analyse many different aspects of our CanSat with one small sensor.

Turbulence and stability is relevant to flights in general as high values of turbulence can often lead to mission failure, or injury if humans are involved. Studying how turbulence affects a smaller system such as a system provides a simplified model for understanding the effects of turbulence on larger models.

This will be done using a MPU6050, a sensor that acts as an accelerometer and gyroscope.

The accelerometer can be used to work out the turbulence, and the gyroscope can be used to evaluate our parachute's stability

We hypothesise that turbulence will decrease with altitude due to reduced thermal convection at higher altitudes, as the influence of heat emitted from the ground decreases

We hypothesise that our parachute will exhibit high stability due to its confident performance in tests.

We also hypothesise increased turbulence will lead to a decrease in parachute stability, showing a negative correlation.

1.2.2 What will you measure, why and how?

For the primary mission:

- Our sensors will be controlled with a Pi Pico
- Air pressure and temperature will be measured with a BMP280 due to experience with the sensor in our group, as well as its ability to read both temperature and air pressure, therefore saving space in our can.
- To send and receive radio signals we will use two Adafruit LoRa radio modules and a Yagi antenna as we are familiar with this equipment and buying new, different modules would be risky and exceed our ideal budget.
- To program the sensors we will be using CircuitPython due to its compatibility with the sensors we have chosen
- We will calculate the terminal velocity of our CanSat by looking at the rate of descent of our altitude measurements.This will be taken only once the rate of decrease has stabilised to ensure accurate estimates.
- We can also calculate the drag coefficient of our parachute. At terminal velocity, the force of drag is equal to the weight force of the CanSat. Then we can do similar calculations to those we used to calculate the size of our parachute to work out the drag coefficient. These calculations from tests have been used to improve our parachute's design.

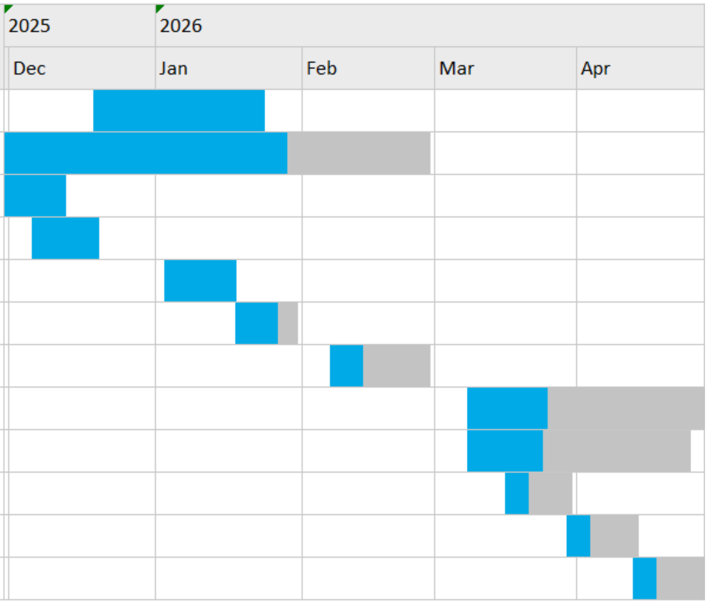
For the secondary mission:

- To calculate the turbulence the accelerometer component of the MPU6050 will be used. We will ignore movement in the vertical axis and only consider movements in the horizontal axis as the vertical axis is largely affected by the rate of descent. A small range of values would indicate a low turbulence, and a large range would indicate high turbulence. By taking the standard deviation of our accelerometer data, we can model turbulence as fluctuations in our CanSat's movement during descent
- The gyroscope element of our MPU6050 returns values in degrees per second. To give a measure of stability, we will take the standard deviation of these measurements, similar to our calculations for turbulence. However, we want to also classify a high average rate of rotation as unstable, and so we will also encapsulate the mean value in our evaluation function.
  A simple evaluation function that we will use is Standard deviation * Mean, as it is a composite measurement that captures both sustained levels of high rotation and random oscillations.
- We will use the Python libraries pandas, numpy and matplotlib to analyse our data after launch. This choice is justified as one of our team members has experience in all three, and due to their plethora of built in functions and data visualisation options. Examples of our data analysis can be found in the last section of 3.6, testing.

2 PROJECT PLANNING

2.1 Time schedule

Tasks	Resp	Status	Start	Due
CDR completion		Done	12/19	1/23
Regional Launch Preparation		Doing	11/30	2/27
Software implementation		Done	11/30	12/12
Electronics assembly		Done	12/6	12/19
CanSat Assembly		Done	1/3	1/17
Testing for launch conditions		Doing	1/18	1/30
Practice launch		Todo	2/7	2/27
FDR completion		Todo	3/8	4/27
National Launch Preparation		Todo	3/8	4/24
Practice Launch 1		Todo	3/16	3/30
Practice launch 2		Todo	3/30	4/13
Data analysis		Todo	4/13	4/27



We have started our Gantt Chart from the PDR submission date. This shows a rough overview of how we have spent the time from the PDR up to the CDR and how we plan to spend our time after this.

We have kept our strategy of splitting many tasks across our group, completing them in parallel to meet deadlines faster and more efficiently. There have been some issues with this. Creating a fully functional CanSat and completing all tests within the time frames we presented in the PDR proved difficult due to team members' limited availability, especially over the Christmas period. However, we are now back on track and will have more than enough time to prepare for the Regional Launch.

For the CDR we split the work between people by giving everyone a section to do. This allowed for quicker completion of documentation, as well as making sure that people could apply their strengths to their respective sections. The CDR completion was about 5 hours, due to the extra detail needed on top of that of the PDR.

For the Regional Launch preparation, we split our tasks into Software, Electronics and CanSat Assembly, as well as some tests for launch conditions and a practice launch.

Software entailed the coding of each sensor in CircuitPython, and our design on how we wish for them to interact. Due to our experience in the competition from last year, this took around 2 hours, as only the MPU 6050 was new to us..

As for electronics we needed to decide how we wished to lay out our circuits, and work on soldering and wiring. This was similar to software in that we had done this last year, but due to limited experience in soldering this took around 3 hours.

Our Cansat's basic assembly is complete, with our electronics and software both tested together. We have printed a prototype and are currently iterating through CAD sketches to improve the design to its best possible state for the launch. However, a higher priority is ensuring that all our electronics work exactly as intended and that the ground station functions excellently at high ranges, as this is what poses a higher risk of failure and is more relevant to the mission of the competition. The assembly of all electronics and software is more important at this stage, and this took around 2 hours, mostly spent on troubleshooting issues with code and wires.

Testing for launch is in progress, with all of our sensors checked for their function. All but the LoRa transceiver function as intended, although we will likely be able to work out this issue in the time before the launch. We also intend to test the parachute using a mass similar to that of the CanSat, to ensure descent occurs at the wanted velocity. These tests are simple yet time consuming, and we estimate that the total expenditure will be 4 hours due to the plethora of necessary tests, and the actions we need to take based on the results of these.

We also plan on conducting a practice launch closer to the date of the regionals to ensure that our fully functioning CanSat is able to perform as we wish, and so that we are better prepared for the actual launch day. This will likely be done using a drone from our mentor, attaching the CanSat to this and raising it to a certain altitude. This will allow us to collect data in the same environment that the can will experience on the launch day, making sure that we collect the intended data, that the data matches our expectations and giving us some real world data to practice analysis on. We believe that this task, which should take 4 hours in total due to the planning beforehand and data analysis afterwards, will let us be far more prepared for the regional launch, minimising our risk of failure on the day.

Should we be selected for the nationals, we will need to prepare for the national launch and complete the FDR. For the FDR we will take a similar approach to the CDR, which is explained above. However, we expect it to be more time-consuming, estimating that it will take around 7 hours in total due to the additional sections and detail in comparison to the CDR.

For the national launch, we plan to conduct two practice launches before the launch, as well as some time to analyse our data properly to present in the nationals. The launches should take 4 hours each, and the data analysis should take 2 hours to analyse and present nicely.

2.2 Team and External Support

	Programming	Electronics	Mechanical design	Outreach and leadership
Student 1 (Team Lead)	Excellent programming skills specifically with micropython	Substantial experience with electronics	Some skills and has good ideas	Lots of experience with public speaking and excellent leadership skills
Student 2	Excellent programming skills, extensive experience with micropython	Substantial practical experience with electronics	Few skills but has good ideas	Some experience with public speaking and good leadership skills
Student 3	Excellent programming skills with microcontrollers	Lots of applied practical experience with electronics	Very good ideas and experience designing	Some experience with public speaking and confident leadership
Student 4	Good programming skills especially with micropython, but may require more support	Some experience with electronics	Excellent ideas and plenty of experience designing for several purposes	Experience with public speaking and leadership
Student 5	Little programming skills, not his area of expertise	Little experience with electronics, but could be taught	Would come up with plenty of ideas and some experience	Experience presenting and some leadership experience
Student 6	Some basic programming skills, requires assistance from peers	Little experience with electronics, but could potentially be taught	Good ideas and some experience	Experience presenting and confident in leadership experience

Best Skill Strong Skill Secure Skill Weak Skill

Overall, we believe that we possess a well-balanced skillset as a team, with strong skills across key areas for the project. These skills map well to our team roles, meaning everyone is in a position well-suited for them. Team members have sufficient skills in areas outside of their expertise to support others in their responsibilities, and we have enough time to teach weaker members so they can also be of assistance.

However, despite our well-balanced skill set there are still some areas we will likely struggle with. Although our team leader is very experienced in outreach, other team members may lack the confidence to run activities such as workshops or school assemblies. To solve this, we will slowly build up to the most challenging activities by doing activities that are less interactive or with smaller groups of people first, allowing students to gain the necessary confidence. We will also work closely with our local robotics club and potentially have other members of the club or staff assist with the activities we are running. Another key area we may require support with is soldering. Although our team has very good experience with robotics, we do not have much experience with soldering. As good soldering is vital to the reliability of our CanSat, as our team's lack of experience could cause major issues. We will need to work on this skill and/or seek external support by getting someone to teach us, likely from our robotics club.

If there are any other areas in which we require support, we will seek it from our mentor, the competition guidelines and friends who have participated in this competition in the past. Our mentor is able to provide most of the resources and knowledge that we need, but we have also found online resources such as those provided by STEM UK to be very helpful.

2.3 Risk Analysis

What follows is a reassessed and updated risk analysis and mitigation plan from the one included in the PDR. Currently, we are less concerned about team management as everyone seems to be comfortable and working effectively in their roles, however the risk of not finishing major sections of the project on time due to poor time management remains as prominent as ever. Additionally, although we have performed extensive testing and attempted to ensure all equipment is reliable, there is still a risk that equipment will fail, so a main goal will be to put in place measures so this absolutely does not happen. There are also new risks to personal health and safety as we are now carrying out more of the practical elements of the project, which are detailed below.

Risk	Consequences of this risk	Current and planned mitigations
Team management difficulties/skill gaps	Could lead to poor documentation and designs if team members are unclear on their task, are assigned to a task they are not comfortable with, or do not have a task to do. This will contribute to the probability of technical failures on the launch day and in general. The team would not be being utilised to its full potential.	Team members have been organised into roles based on their skills, and which have proven effective thus far. However, it will still need to be ensured that all members have clear goals at all times, especially when working on critical documentation, so that the workload is spread evenly. The same organisation will also need to be applied to the practical building of the CanSat to ensure everything is built as reliably as possible and in line with the minimum guidelines.

Time constraints	Possibility of unfinished or poor quality documentation as deadlines are not being met, critical oversights in design leading to preventable failures of the CanSat, or unfinished assembly of the CanSat before launch day, which has affected teams in previous years.	Currently, we are somewhat behind the schedule that was laid out in the PDR, however we believe we are currently on track to catch up. The reassessed time schedule included in the CDR includes detailed, strict plans on when all sections of the project must be completed by. Our plans also ensure that most aspects are completed long before they are actually needed to be so, giving ample time for testing and dealing with unforeseen challenges.
Equipment failures	Unreliable equipment at the ground station ,such as a poorly assembled antenna, or issues like poor soldering inside the CanSat itself, will likely cause us to lose some or all of our data, negating much of the work put in and giving a disappointing outcome to the launch.	Faulty components may be the result of some issues, and these can be replaced or we can attempt to fix these, seeking help from our mentor if needed. It may also be possible to request a replacement component from our mentor. We may also minimise the risk of faulty components by carrying out soldering in a calm fashion and ensuring it is a task only given to experienced team members
Health and safety	Fumes from solder can be very dangerous if exposed to for prolonged periods of time while building the CanSat, as well as risks of burns from the soldering iron itself. Additionally when drop testing the CanSat from a high place it can cause serious injuries to both team members and the public below.	To minimise the risk of personal harm, all soldering will be done in the safest possible environment, with anti-static heatproof matting and ideally fume hoods. We ensure the area where soldering is taking place to be calm and controlled, and that the person soldering knows how to do so correctly and safely. During drop tests, we will minimise risks by performing them in quiet areas with few or no people nearby, and ensuring all people in the area know exactly when and where the test is happening so they can avoid the testing area.

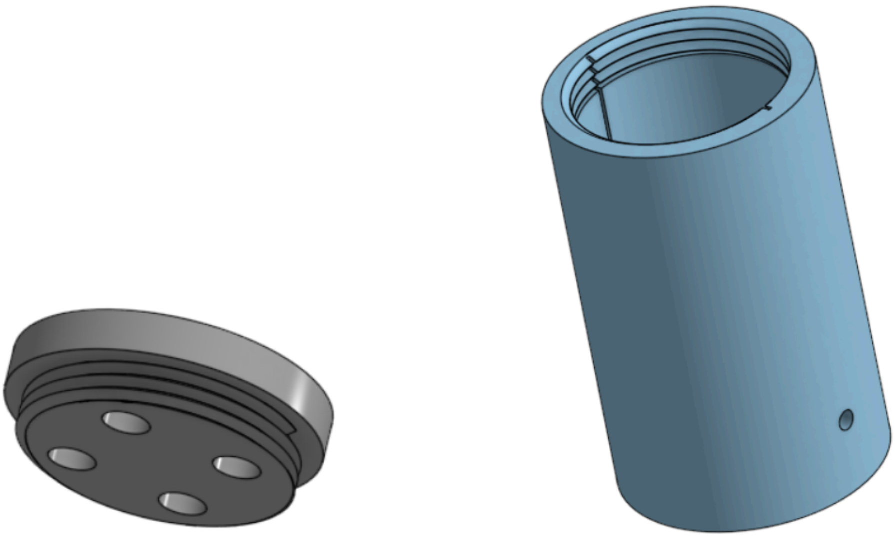
3 CANSAT DESIGN

3.1 Mechanical design

The above photo shows an iteration of the CanSat, which we used to test the effectiveness of airflow holes. Due to an error on the CAD, the screw mechanism was not fully functional, however this issue is rectified on the Final Design. We have also decided not to include airflow holes due to the lack of benefits it provides, as well as to increase our CanSat's structural integrity.

Final Design

The final CanSat includes a body of 106mm and a lid of 18mm, which overlap by 9mm for the screw mechanism. Each of them have a 66mm diameter, and therefore meet the given size requirements (66mm x 115mm). On the lid, there are four holes of 10mm, for the attachment of a parachute to the CanSat, as planned in the PDR. However, we opted not to include holes in the sides and bottom for airflow, as in prototyping we found that they did not affect the sensors' results, but did compromise strength. There remains one 5mm circular hole on the lower half of the body, which is for the antenna to communicate with the ground. To attach the lid to the body, there is a screw mechanism, as it is strong, simple, and easy to manufacture. We chose this above the alternatives described in the PDR as it has proved strongest and most reliable in testing. Inside of the body there are rails on opposite sides of the walls, which will allow for the protoboard and all the attached components to slot inside securely. We have chosen to 3D print the CanSat, as we have one available within our team. Although there may be durability issues on the vertical axis due to the build direction of our 3D print, this should not be a major issue as there will not be pulling force across this axis. We have chosen to print in ABS due to its durability, as well as its high impact and weather resistance. Strength is key for the CanSat due to the extreme forces of up to 30G exerted onto it during the launch. However, in prototyping we have used PLA, as it is cheaper and less toxic.



3.2 Electrical design

Following on from the PDR, we have now finalised our choices of sensors, making only minor changes from our initial plans. This is because we have already obtained and tested most of our sensors, and as results have concluded that there are currently no issues, we see no reason to change our plans in any major way. Detailed justifications for our sensor choices are as follows, with accreditations for the statistics used included in appendix A:

Raspberry Pi PIco

For our main microcontroller, we have chosen the Pi Pico. Although other microcontrollers such as the Arduino Nano or Mini could provide a smaller form factor and lower current draw, we have tested that there is ample space and sufficient power for a pico, and therefore it is undoubtedly the superior option. The advantages of being able to use a pico include:

- Simpler programming with Python, making it much easier to teach less experienced team members and run outreach workshops
- Larger size means more interfaces to wire sensors to and therefore more freedom in design
- Native compatibility with Adafruit’s circuit python which all sensors are fully compatible with
- Current draw of 150mA, making it easy to meet the minimum battery life required with a sufficient battery
- Fast two-core M0 processor

BMP 280

The BMP 280 is a compact sensor to complete the primary mission goals of measuring air pressure and temperature. We chose a BMP sensor as they have a small form factor and low cost, while still remaining reliable and accurate and being well known and compatible with our microcontroller. The BMP 280 specifically has a number of advantages of over other models including:

- Very low form factor
- Low voltage and current requirements, consuming only 2.7μA
- Low cost, and some team members already have this available
- Operating temperatures between -40 and +85 °C, easily meeting mission requirements
- Operates on a simple I2C interface
- High pressure reading accuracy of ±0.12 hPa, equivalent to ±1 m at 25C
- Great compatibility with adafruit libraries

MPU 6050

The MPU 6050 completes our secondary mission of calculating turbulence by measuring both the acceleration and rotation of the CanSat in one, compact package. Although accuracy is not high in readings, the MPU is unbeaten in size and cost, and it is therefore the best choice considering the limited space of the CanSat and the fact that high accuracy is not critical for our purposes. The advantages of our choice include:

- 20mmx15mm form factor
- Operated a 3.3V and only requires 3.9mA of current
- Can be calibrated to accelerations of up to ±16g, and rotations of ±2000°/sec
- Low cost
- Operates on a fast 400kHz I2C interface
- Great compatibility with adafruit libraries

RFM9x LoRa

For our telemetry, we have chosen the RFM9x LoRa from Adafruit. With the correct antenna setup and conditions, the RFM9x can have a range of up to 2Km, ideal for our CanSat, and overall the best range that can be achieved with a module of its size and price range. The module is much more expensive than our other sensors however, but by splitting the cost this is manageable. For the antenna, the simplest and most space effective solution is to solder a quarter wave length wire antenna, to achieve the maximum possible transmission power. As the radio operates at 433Mhz, and wavelength = speed/frequency, the wavelength ≈ 0.69m, so the antenna will be 0.17m long.

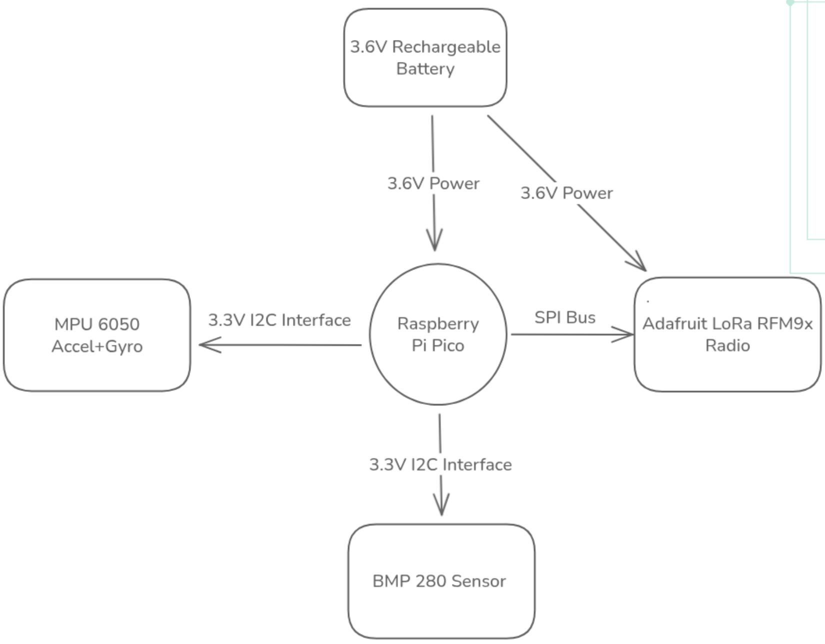
- 2km maximum range
- Peak transmission current draw of 200mA
- Data sent in 255 byte packets
- Fast data transfer over an SPI bus
- Native compatibility with adafruit libraries

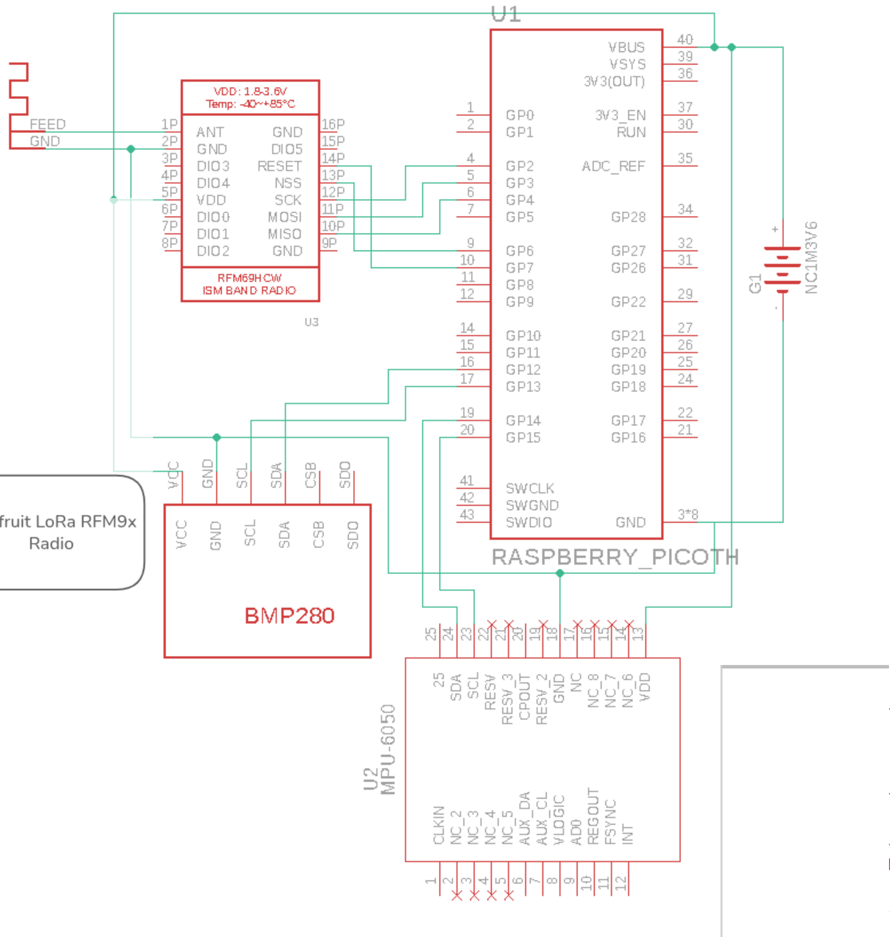
Power

To power our CanSat, we have chosen a 3.6V Li-Ion rechargeable battery. Although 5V would be ideal by using 3 AAA batteries, this option has been tested to give up to 4.2V, and the space restrictions of the CanSat limit us to smaller options with lower voltages regardless. he battery will give a minimum of 800 mA hours, and as the Pico only requires 150 mA, that is a minimum of 5.3 hours of battery life, surpassing the minimum requirement of 4 hours. Additionally, the battery can be recharged easily via micro USB over 2000 times, meaning the battery will never need to be replaced and is easily accessible.

Circuit and System Diagrams

The following is a high level system diagram showing the overall flow of data and power between components. The MPU 6050 and BMP 280 are powered at 3.3V from the Pico as they can operate at full efficiency at this voltage and need very little current, however the radio is powered directly from the battery due to it requiring the closest voltage possible to 5V for optimal transmission power. The radio is also on an SPI bus due to the large volume of data needed to be passed over, while the sensors are an I2C interface.The following the system diagram is a more detailed schematic of how each wire will be soldered and roughly how the components will fit together inside the CanSat.





3.3 Software design

Overall code design

Our programming language of choice for our CanSat is CircuitPython. As mentioned before hand in section 3.2, CircuitPython offers excellent compatibility with our chosen sensors and microcontroller, while still being relatively easy to program, and is an exceptional language for teaching others using the resources we have developed. The ground station will be programmed in CircuitPython along with the actual CanSat. The data collected will then be sent over a serial connection to a laptop running a program to visualise and save this data to a csv file, also written in python. This will use the Tkinter and serial libraries to receive and display the data in a simple but clear UI. Finally, the necessary calculations for our secondary mission goal as well as the processing and visualisation of the data will be done in an iPython notebook using Matplotlib and Pandas. More details on the data processing is included in section 3.5.

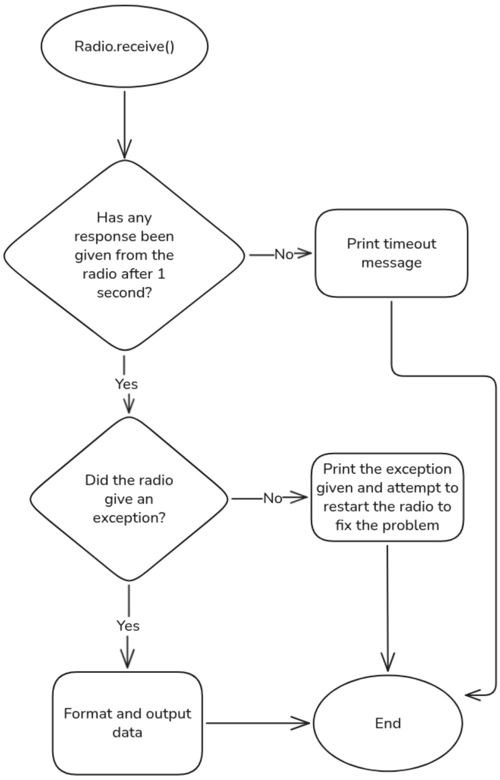
The code for our CanSat revolves around a single library that contains custom classes to interface with the main hardware components of the CanSat. This library achieves a number of important goals:

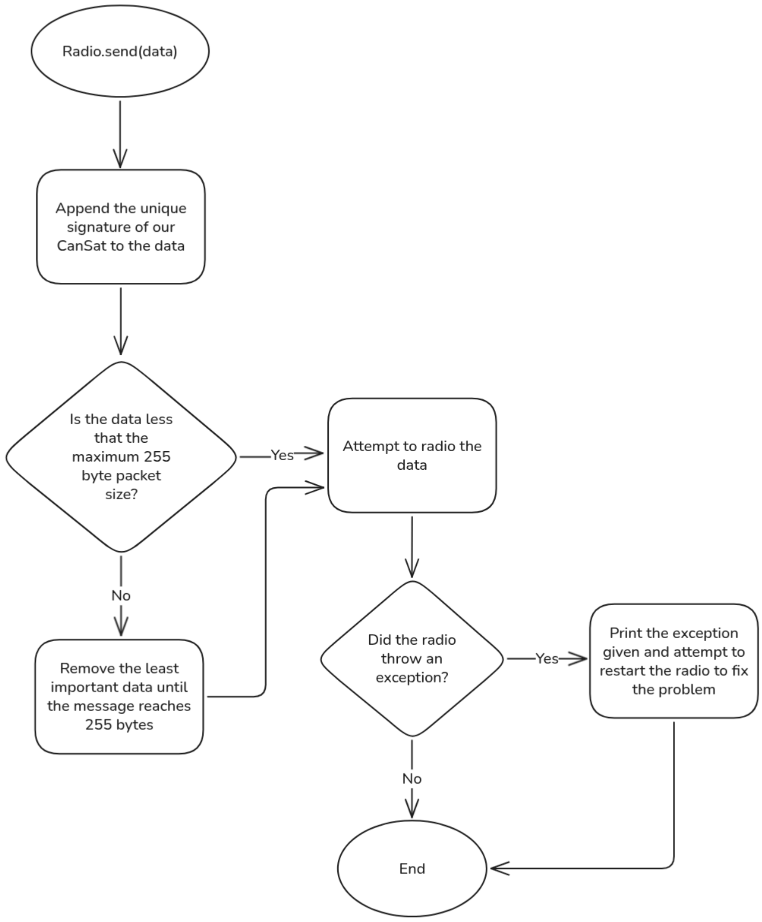
1. The library abstracts the somewhat complex process of interfacing with the hardware, making the end code very simple to understand.
2. The simple to use nature of the library gives us excellent resources for outreach workshops.
3. The library decomposed the code down into more manageable chunks, making code more readable and maintainable.
4. As the file is shared between both the ground station and actual CanSat, both are always guaranteed to be compatible and have the latest version.

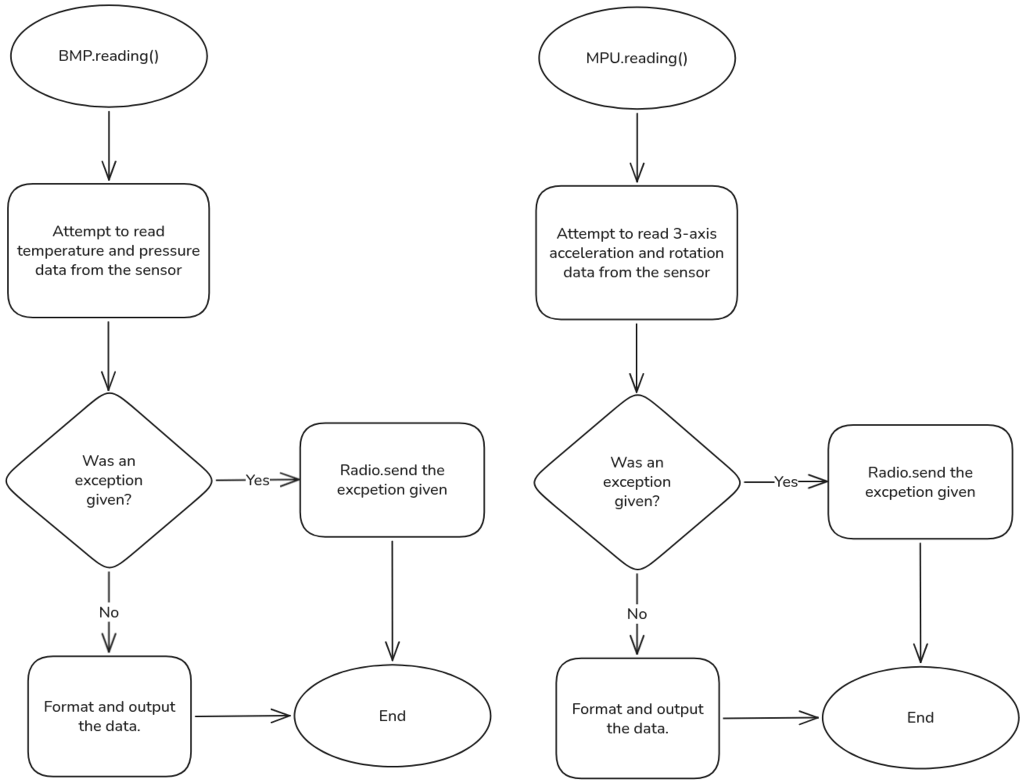
Around this we have built very simple programs that can be easily modified at any time and have last minute adjustments made. As reliability is critical for applications such as CanSat, we have designed our software around the idea of minimalising all potential crashes and bugs. CircuitPython will restart the entire CanSat upon an exception being thrown automatically, and this would mean a loss of most of our launch data. Therefore, the code will catch all exceptions, attempt to radio the error to the ground station, and restart the necessary sensors to attempt to fix the problem. Using this approach, a good balance of reliability and ease of use is achieved.

Library code

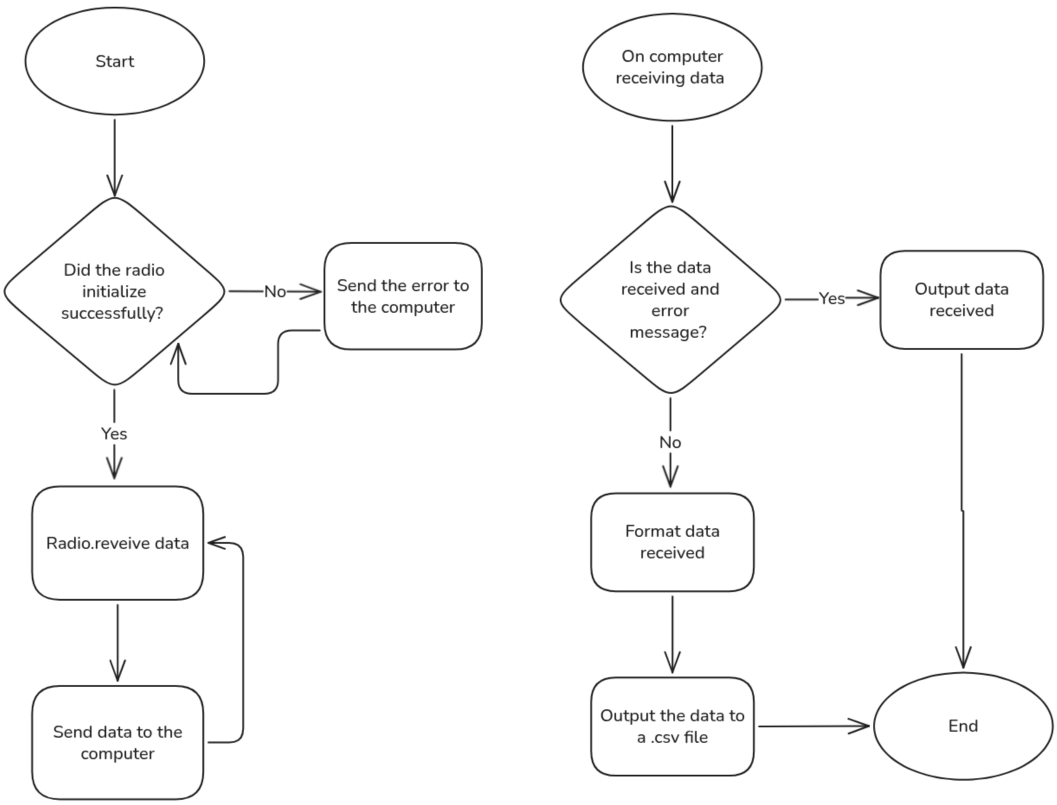
What follows are flowcharts for all the functions of our library. As the radio is slightly difficult to interface with, and it is the most critical component, extra care has been taken to omit any errors with this section of the program. The sensors' programs are much simpler. After the library code is the code for the final CanSat and ground station based on this.


Radio.receive()
Has any response been given from the radio after 1 second?
No
Print timeout message
Yes
Did the radio give an exception?
No
Print the exception given and attempt to restart the radio to fix the problem
Yes
Format and output data
End

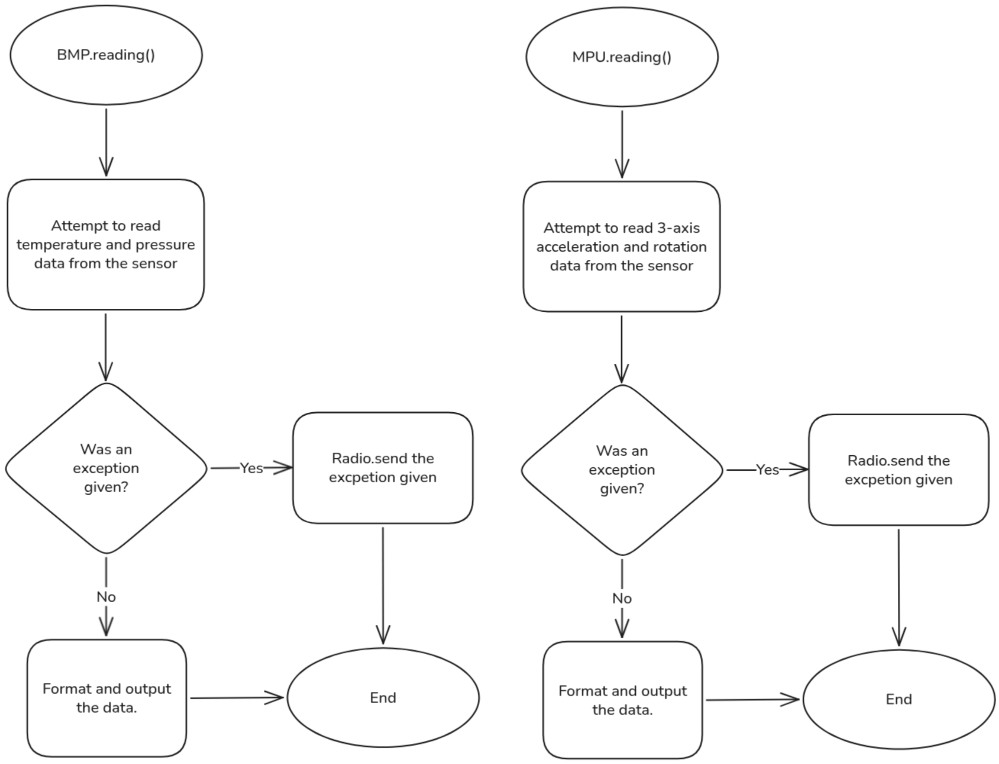

Radio.send(data)
Append the unique signature of our CanSat to the data
Is the data less that the maximum 255 byte packet size?
Yes
Attempt to radio the data
No
Remove the least important data until the message reaches 255 bytes
Did the radio throw an exception?
Yes
Print the exception given and attempt to restart the radio to fix the problem
No
End



Ground station code



CanSat code



3.4 Landing and recovery system

Parachute Design:

For the CanSat's landing system, possible parachute designs included a paragliding design, a flat hexagonal design, a hemi-spherical design or a cross parachute design.

Paragliding design:

- This design offers a drag coefficient of ~0.75 to 1.1, this is an effective value for our CanSat
- This design is suitable for a guided descent, but we do not require this
- The design is difficult to be made accurately by hand, may cause unexpected errors during launch

Flat hexagonal design:

- This design offers a drag coefficient of ~0.75 to 0.8, this is an effective value for our CanSat
- The design would benefit from a spill hole for stability
- The design can be made accurately by hand

Hemi-spherical design:

- This design offers a drag coefficient of ~0.62 to 0.77, this is an effective value for our CanSat if we make it with high accuracy, too low of a drag coefficient would increase the cross-sectional area, potentially causing issues during deployment
- The design offers high stability and would benefit from a spill hole
- The design is extremely difficult to make accurately by hand

Cross parachute design:

- This design offers a drag coefficient of ~0.6 to 0.8, this is an effective value for our CanSat if we make it with high accuracy, too low of a drag coefficient would increase the cross-sectional area, potentially causing issues during deployment
- The design can be made accurately by hand

Having considered all these points, we have decided to use the flat hexagonal design for our parachute. This is largely due to its high drag coefficient and being able to be made accurately by hand. We chose this design over the cross design because of less potential errors during deployment caused by the hexagon's higher drag coefficient.

Parachute Sizes:

To find the cross-sectional area for the parachute, we need to consider the following constraints:

- The descent rate of the CanSat must be around 10 m/s
- The CanSat's mass must be between 300 - 350 grams
- For this calculation, as this is only an estimation, we will be taking the upper bound for the mass

For this calculation, we will be using this equation which expresses the equal weight force and drag force when the terminal velocity is reached:

$m \cdot g - \frac{1}{2} \cdot C_D \cdot \varrho \cdot A \cdot v^2 = 0$

The variables in the equation are:

m = mass of CanSat, in this case, ~350 grams; g = gravitational acceleration on Earth, ~9.8 m/s^2; C_D = drag coefficient, the total drag resistance, not directional, in this case, 0.75, taken from the lower bound in the case of slight inaccuracies in the making of the parachute; ϱ = fluid density, in this case, density of air (however only at sea level, provides a moderately valid estimation), 1.225 kg/m^3; A = cross-sectional area of the parachute; v = the required terminal velocity of the CanSat, in this case, 10 m/s.

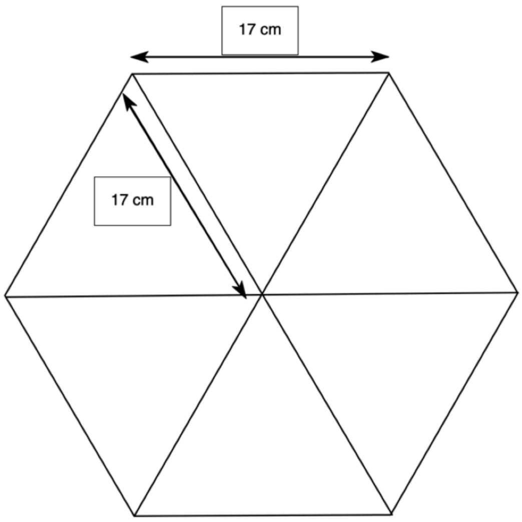
This can be rearranged to make area the subject:

$A = 2 \cdot m \cdot g / C_D \cdot \varrho \cdot v^2$

Now we substitute the values:

A = $2 \cdot 0.35 \cdot 9.8 / 0.75 \cdot 1.225 \cdot 10\text{^}2 \approx$ 0.0747 m^2 or 747 cm^2

From these calculations we reach the parachute design of a flat hexagonal shape with a side length of ~17 cm and a cross-sectional area of ~747 cm^2.



An image of the completed parachute is included in section 3.8, proof of work.

Spill Hole:

For ensuring the stability of the descent, we will be adding a spill hole to the parachute. This is for preventing the parachute from catching air and expelling air irregularly at the sides, causing stability issues with swaying. We decided to choose between the spill hole sizes of 10%, 15%, and 20% of the parachute's cross-sectional area. We will check which percentage will be effective, while still having a reasonable safety margin for velocity with the decreased cross-sectional area.

Rearranging the equation for terminal velocity gives:

v = $\sqrt{2 \cdot m \cdot g / C_D \cdot \rho \cdot A}$

Substituting the values for 10%:

v = $\sqrt{2 \cdot 0.35 \cdot 9.8 / 0.75 \cdot 1.225 \cdot 0.06723} \approx$ 10.54 m/s

Substituting the values for 15%:

v = $\sqrt{2 \cdot 0.35 \cdot 9.8 / 0.75 \cdot 1.225 \cdot 0.063495} \approx$ 10.84 m/s

Substituting the values for 20%:

v = $\sqrt{2 \cdot 0.35 \cdot 9.8 / 0.75 \cdot 1.225 \cdot 0.05976} \approx 11.18$ m/s

Having considered all these values, we decided to for a spill hole with an area 10% of the cross-sectional area of the parachute (or ~ 74.7 cm^2) cut at the top of the parachute. This is because it provides stability while still maintaining a reasonable descent speed assuming the drag coefficient is 0.75.

Shroud Lines:

Now for the shroud lines we had chosen for its length to be 1.15x the diameter of the hexagon from one corner to the other (34 cm). This is a common length for CanSat shroud lines and for standard parachutes. $1.15 \cdot 34 = 39.1$, for ease of practicality we will be using 40 cm. For connecting the shroud lines to the CanSat, we used a nylon cord also 1.15x the diameter (40 cm). This is for ensuring strength and durability during deployment.

3.5 Ground support equipment

Our ground support equipment is essential for the function of our CanSat, as it ensures the transfer of data from our CanSat to the ground and allows us to ensure the smooth function of our mission. Our main setup will involve a Yagi Antenna combined with the Adafruit RFM9x LoRa Transceiver, as this allows us to communicate with our CanSat, as well as a laptop. However, we will use other equipment to ensure the smooth preparation for our launch.

On the ground, the most important component for us is the antenna. As mentioned in the electrical design section, weak radio connection is a major risk. Therefore, for our antenna we will use an Alliance UHF yagi antenna. This antenna is perfect as our radio modules operate at 433Mhz in the UHF band, it is very lightweight and manoeuvrable, and operates with a high gain of 7 dBi. While other antennas could be used, this antenna provides good gain and power while being in the lower price range. However, the antenna is uni-directional, so we will have to track the visual trajectory of the CanSat in order to get optimal results. We will dedicate a person to this on the launch day, so it should not be an issue, but this still may reduce the efficiency of the radio as it will be very difficult to track it at high altitudes.

Another very important thing to consider is the radio. We will use two LoRa transceiver modules, one in the can, fitted with a small wire to act as a basic antenna, and one on the ground, which we will combine with our directional Yagi antenna, as using two identical radios will always ensure compatibility. These will be connected to a Pi Pico and the laptop via a data transmitting USB file, which will allow us to directly record data into a CSV file using custom software in Python. The Yagi antenna requires precise alignment with our CanSat, so during descent we will assign one team member to spot the CanSat and another to direct the antenna. This will ensure that we get a constant stream of data, minimising the work we need to do later on in order to clean up and analyse our data.

As well as the equipment necessary during the launch, we will take some to prepare for it on the day. A soldering iron is necessary to ensure that we are able to carry out any needed last minute adjustments to our wiring. We will bring ample spare components to ensure any last minute electrical breakages can be fixed, as well as hot glue and strong tape to fix last minute mechanical breakages. Although this should not happen, it is critical to prepare for all situations. Additionally, we will bring multiple laptops with the necessary software installed as well as backup power supplies.

3.6 Testing

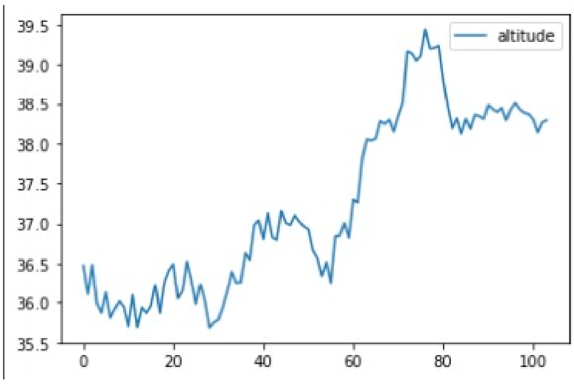
What we tested	How we tested	Initial testing results
Structural integrity	Repeatedly drop the CanSat from a height of around 1m. Inspect for loose parts and broken connections. Check if telemetry is functioning before and after the test	The CanSat is fully intact after being dropped, even on hard surfaces.
Size and weight restrictions	Measure the dimensions and mass of our CanSat to ensure that it is within the required range	Our can has been designed using CAD software, ensuring that our can meets the size dimensions required by the competition guidelines. We have also measured its mass with all components inside, and it weighs slightly under the 300-350g range, so we will likely add ballast that can also function as cushioning for components.
Battery accessibility	Time how long it takes for someone to turn off the battery with gloves on	The battery will be soldered at the very top of the CanSat and can be very easily removed from the holder.
Battery life	Run the CanSat with the fully functioning sensor for 3 hours and observe results	The CanSat will run for over 4 hours with all sensors functioning.
BMP-280 temperature accuracy*	Place the BMP in front of a hairdryer and in a freezer, recording how close it was to the actual temperature of those sources.	Our testing results concluded that the BMP reflects the temperature of its environment with the accuracy required, therefore no action is required

What we tested	How we tested	Initial testing results
BMP-280 pressure and altitude readings*	Use the pressure readings to calculate altitude values and see if they are reasonable and in line with reference values. Check if altitude reading increases at the top of a tall building	Our sensor's readings of pressure are in line with those expected for our area. As well as this, the altitude reading calculated from our sensor increases visibly at the top of a building compared to the bottom, in line with our expectations, so no action is required.
MPU 6050 accuracy*	Record values when stationary and when moving, and did the same with orientation by rotating the sensor in specific axes	The sensor detects acceleration and rotation in 3 axes clearly, however accuracy is low at the speeds we tested at. However, at the high speeds of the CanSat during launch, the sensor should be more accurate, but further testing may be required.
Radio functionality	Run the radio connected to all of the sensors and observe the output from telemetry and if packets are intact.	Packets are received consistently and intact at very short range, but issues arise at longer range tests, as described below.
Radio range	We attempted to check if the radio is fully functional at a range of at least 1km	Data cannot be received at ranges greater than 20m. The issue is very likely to have been caused by an unreliable antenna or radio module, so further testing will include replacing these components, see below
Parachute length	Test if the parachute connection can withstand a force of 50N.	Confident parachute deployment
Parachute reliability	Drop the parachute from a reasonable height repeatedly and confirm that it deploys consistently	Parachute deploys reliably with only a little weight required. A 10% area spill hole may however be larger than necessary so further testing will be undertaken to inform our decisions.
Descent rate	Drop the parachute from a reasonable height and check	Requirement met

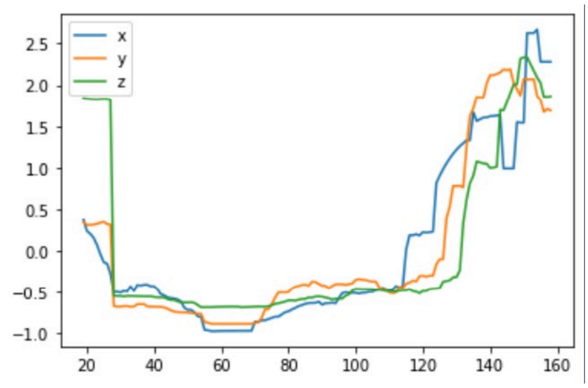
What we tested	How we tested	Initial testing results
	if its terminal velocity is within the required limits	
Descent stability	Observe the CanSat during descent and ensure that there is not excessive oscillation	Valid secondary mission data, safe descent

*The results of these tests are described in the section below

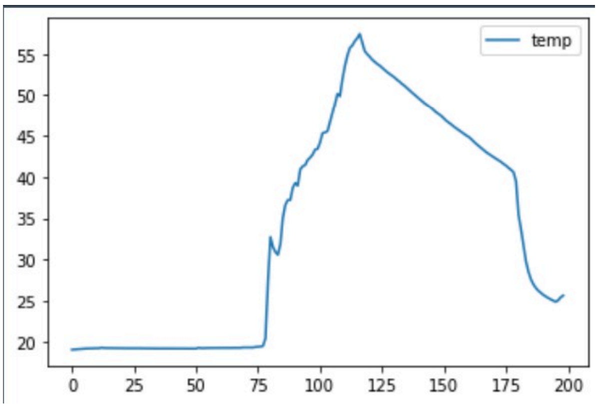
Overall, our testing has proven that most areas of the CanSat work effectively and in line with the guidelines. Two sections that may require further testing are the MPU6050 accuracy and the parachute reliability, as these are critical for mission success and showed weakness in testing. However, they both still function well and therefore further testing will be done when the full CanSat is assembled, and adjustments can be made from there. A much larger priority for testing was the radio. We managed to pinpoint the problem as an unreliable antenna, and have since replaced the antenna and seen much more reliable results up to 2km.



Graph 1: Altitude readings obtained from pressure values when going up one flight of stairs, shows BMP280 is accurate within a tolerance of +-0.5m



Graph 2: Turbulence values in all 3 axis calculated from accelerometer readings when running in a straight line and then shaken



Graph 3: Shows the BMP being heated with a hairdryer for 30 seconds, cooled slowly and then cooled rapidly by placing a finger on it

3.7 Overall testing for launch

Once the CanSat has been fully assembled and all of the tests detailed above have passed, we will attempt a full emulation of a CanSat launch using our mentor’s drone. The process will go as follows:

1. We will set up all ground station equipment, and assign members to the roles that they will carry out in the actual launch.
2. The CanSat will be fastened to the drone, including parachute, and rigorously tested to ensure it is safe.
3. The CanSat will be taken up to the maximum height of around 100m, and the ground station will be observed.
4. The drone will drop the CanSat and the effectiveness of the parachute will be monitored closely as possible for any issues.
5. We will recover the CanSat from its landing site.
6. The data collected will be analysed as if it was actually launch data.

Other tests related to the launch have been covered in the testing section

3.8 Evidence of CanSat build

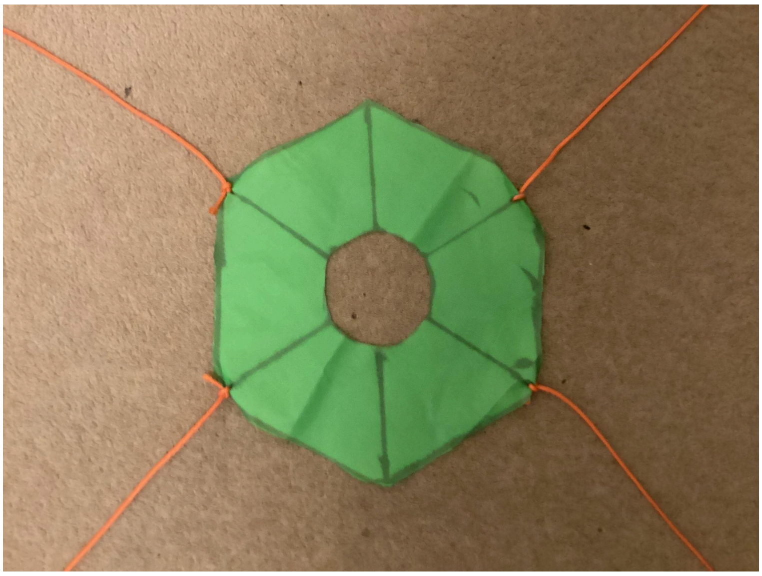


Image 1: A photo of our parachute

Image 2: Our complete 3D printed design

4 OUTREACH PROGRAMME

For outreach, we have taken a wide approach, using various methods to engage many different demographics

Our main outreach achievement is the confirmation of a section in our local newspaper for after the regional launch. This newspaper delivers to 5000 homes and businesses free of charge monthly, and is read by a wide range of demographics. By advertising the competition in such a newspaper, we can both encourage peers of our age to participate in such STEM-related activities, as well as raising awareness of the competition for adults. It also allows children who do not have access to social media to be more aware of such competitions. We have also created a blog for our CanSat, detailing the nature of the competition as well as documenting our progress over time. This will be displayed via our mentor's robotics club, which has engaged around 1200 students locally. By posting on social media, we can appeal to both adults and peers of our age, maximising the impact of our outreach.Our final outreach activity is to host a workshop showcasing our CanSat and raising awareness of such STEM competitions. This will also be conducted via our mentor's robotics club, which has vast experience in running such events. The main benefit of such outreach methods is that it appeals directly to those with an interest in STEM, and is incredibly engaging, encouraging others to participate in such competitions.

APPENDIX A

The accreditations for the statistics given in section 3.2 electrical design are as follows:

BMP 280 -
https://cdn-shop.adafruit.com/datasheets/BST-BMP280-DS001-11.pdf

MPU 6050 -
https://invensense.tdk.com/wp-content/uploads/2015/02/MPU-6000-Datasheet1.pdf

RFM9x LoRa -
https://cdn-learn.adafruit.com/assets/assets/000/031/659/original/RFM95_96_97_98W.pdf?1460518717

Pi Pico -
https://pip-assets.raspberrypi.com/categories/610-raspberry-pi-pico/documents/RP-008307-DS-1-pico-datasheet.pdf?disposition=inline